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Predictive analysis of heart disease using quantum-assisted machine learning

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Received: 21 October 2024 / Accepted: 14 April 2025
Published online: 03 May 2025
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Abstract
Coronary heart disease (CHD) is a severe cardiac disease, and hence, its early diagnosis is essential as it improves treatment results and saves money on medical care. The prevailing development of quantum computing and machine learning (ML) technologies may bring practical improvement to the performance of CHD diagnosis. Quantum machine learning (QML) is receiving tremendous interest in various disciplines due to its higher performance and capabilities. Techniques for QML have the potential to forecast cardiac disease and help in early detection. To predict the risk of coronary heart disease, a hybrid approach utilising an ensemble machine learning model based on QML classifiers is presented in this paper. Our approach, with its unique ability to address multidimensional healthcare data, reassures the method's robustness by fusing quantum and classical ML algorithms in a multi-step inferential framework. Reducing cardiac morbidity and mortality requires early detection of heart disease. In this research, a hybrid approach utilises techniques with quantum computing capabilities to tackle complex problems that are not amenable to conventional ML algorithms and to minimise computational expenses. The proposed method has been developed in the Raspberry Pi 4B Graphics Processing Unit (GPU) platform and tested on a broad dataset that integrates clinical and imaging data from patients suffering from CHD and healthy controls. The proposed research is developed with a hybrid approach that combines different machine learning algorithms, such as KNN + RF, DT + RF, LR + RF, and Adaboost + RF, for diagnosing coronary illness with higher accuracy through feature selection. The proposed system performance obtained an accuracy of 99%, utilising 20000 datasets with 14 attributes from various datasets collected from the Local Pathology Lab in the Muzaffarnagar District of Uttar Pradesh, India. Compared to classical machine learning models, the accuracy, sensitivity, F1 score, and specificity of the proposed hybrid QML model used with CHD are manifold higher.

Article Highlights

- This research points towards a hybrid QML model revolutionizing healthcare and envisions a future of personalized, predictive treatment frameworks that can address existing and future health challenges.
- This hybrid model effectively deals with noisy and incomplete data by integrating the quantum and classical components. It identifies important features, reduces irrelevant noise, and personalizes the predictions for patient-specific treatment.

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- The Raspberry Pi 5's GPU platform enhances training operations through parallel processing, providing an economical, scalable, and energy-efficient approach to implementing the model in medical devices and remote healthcare environments. It acts as a liquid tool for clinical decision-making and proactive healthcare strategies.

Keywords Healthcare 4.0 · Heart disease · Deep learning · GPU · Quantum machine learning · Quantum neural networks · Quantum bits · Quantum annealing

1 Introduction

Quantum computing and machine learning are two rapidly evolving fields that, when combined, can transform healthcare, particularly in early cardiac disease prediction. Healthcare 4.0 brought about a shift in lifestyle that contributed to the development of heart-related diseases such as diabetes, high cholesterol, obesity, and physical inactivity. Heart disease was the cause of 18 million deaths in 2017, and by 2050, it is expected to account for almost 58 million fatalities [1]. An international concern is the rising frequency of cardiac deaths; according to WHO, 85% of cardiovascular patients reside in low or middle-income nations, and cardiovascular disorders account for 52% of deaths in wealthy nations [2]. Furthermore, it raises the chance of Alzheimer's disease, dementia, COVID-18, and cognitive dysfunction [3–6]. Early identification lowers the risk of these diseases, dramatically reduces the heart disease-related death rate, and enhances the health budget. As a result, in recent decades, early diagnosis of cardiovascular illnesses has been a significant research challenge.

Early detection of coronary heart disease can significantly improve prognosis by enabling earlier intervention and management, which can reduce the risk of severe outcomes such as heart attacks and heart failure. S.P. Patro et al. [7] designed a heart disease prediction framework using five ML models with a UCI machine repository. Hossain et al. [8] propose a hybrid convolutional neural network (CNN) and long short-term memory (LSTM) deep learning (DL) model to identify cardiovascular diseases (CVD) from clinical data combined with feature engineering and explainable AI. However, traditional ML and deep learning (DL) models for predicting heart disease often need help handling imbalanced datasets. Farah Mohammad et al. [9] developed a continuous wavelet transformation and CNN-based hybrid model for Heart Disease Prediction. Their WT-CNN uses RUSBoost-based data balancing and achieves an exceptional accuracy of 97.2% in predicting heart disease. S V Babu et al. [10] proposed a QuEML model for predicting heart diseases using the Kaggle data repository, which performed well in accuracy and computational time. Hybrid QML combines classical ML techniques with quantum computing to uncover patterns in data that traditional approaches might miss. This could lead to more precise and early detection methods for CHD. Quantum-enhanced machine learning is vital in sustaining patient-oriented attention to healthcare patrons. Yogesh Kumar et al. [11] presented a QML model of its significance in heart failure detection on a dataset of 14 attributes. They verified that QRF, QKNN, QDT, and QGNB are better than traditional ML algorithms in heart failure detection.

Hybrid QML techniques could enhance the ability to predict individual risk profiles, allowing for more personalized and effective management strategies. On the other hand, traditional machine learning methods frequently have significant calculation times and little processing power. Quantum machine-learning techniques have been created to address these deficiencies. Superposition and entanglement, two quantum mechanics concepts, are the foundation upon which the QML algorithms are built. An algorithm may assess many positions' impacts on the system response at once by considering the superposition characteristic. As a result, it dramatically reduces computing time. According to the entanglement characteristic, two particles are connected even when they are very far apart. As a result, QML algorithms are appropriate for large-scale or complicated issues and can significantly reduce computing time. The proposed QML methodology has numerous significant benefits over classic Machine Learning (ML) approaches for solving the computational issues of CHD diagnosis. The hybrid QML technique, which blends quantum computing's parallelism and exponential speedup capabilities with classical ML models, has the potential to improve computational efficiency significantly. Quantum algorithms, such as those for data encoding and optimisation, may process vast volumes of data more quickly than classical algorithms because they use quantum superposition and entanglement, which can handle multidimensional feature spaces more effectively. Furthermore, hybrid approaches enable the seamless integration of conventional with quantum algorithms, maximising both classical and quantum resources to overcome traditional ML's inherent scalability and efficiency limitations. QML's low computing cost, particularly in complicated predicting tasks, makes it an attractive option for real-time, large-scale medical applications like CHD diagnosis, where speed, precision,

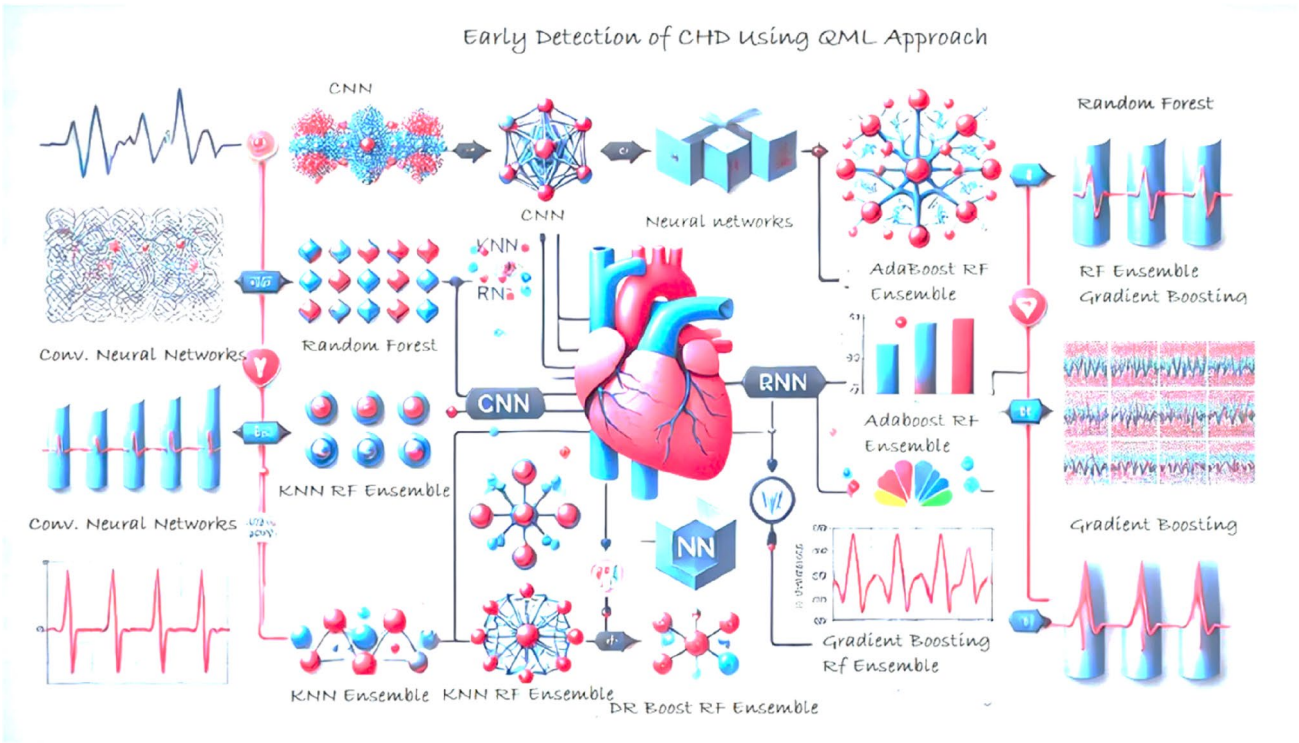


Fig. 1 The proposed architecture of Early Detection of Coronary Heart Disease Using a Hybrid Quantum Machine Learning Approach

and handling massive datasets are crucial. This research suggests a hybrid approach using random forests and more machine learning algorithms like DT, KNN, and Adaboost to address the problem. Despite the widespread usage of QML algorithms and the critical relevance of early coronary artery disease prediction, only some methods currently employ QML algorithms for cardiovascular disease prediction. The proposed QML architecture is shown in Fig. 1.

Figure 1 shows a hybrid and multi-model approach for detecting coronary heart disease using classical ML techniques, including CNN, RNN, Random Forest, KNN, and ensemble learning-based techniques such as Adaboost and Gradient Boosting. A QML integration will significantly enhance computational efficiency and accuracy while operating with large, complex biomedical data, such as ECG signals. Electrocardiogram signals are the main feed of data in the CHD detection process and are fed to several machine-learning models. Convolutional Neural Networks, such as ECG, form one of the most typically used types for time series data. It can effectively learn patterns, local structures, and anomalies from time-series data of heart rhythms. After adding several layers of CNNs sequentially, the outputs are processed through other machine learning models, such as Random Forest and KNN RF Ensembles. Further, the data is passed from CNN into the neural networks for further feature extraction and classification. The neural networks analyze complex nonlinear relationships of features that may signal early appearances of coronary heart disease. Similarly, KNN and RF ensembles were utilized for pattern analysis and to provide predictive insights based on similarities in patient data. Ensembles combine many different models to give better performance with predictability and robustness than one single model alone. Adaboost RF Ensemble uses an adaptive boosting algorithm that focuses on improving the performance of weak classifiers by paying special attention to misclassified cases. It becomes the most potent ensemble for enhancing predictive accuracy with random forest and gradient boosting. Like other ensemble methods to optimize the model's performance, gradient boosting improves the model's ability to make better predictions by taking care of previous iteration errors. Further, these boosting techniques will refine the predictive results developed by the earlier models. The final prediction of these two models would correspond to a high or low risk for coronary heart disease in the patient. After several processing layers, these output results combine the best merits of different models: CNN, RNN, Random Forest, KNN, Adaboost, and Gradient Boosting. It may be applied in areas not explicitly drawn, whereby quantum machine learning models are being brought forward to enhance the speed of computation and the performance of the models.

Healthcare applications must manage sensitive patient data, and QML, as a primary user interface language, frequently struggles to integrate easily with secure backend systems while maintaining data privacy. Furthermore, healthcare systems frequently demand high reliability, scalability, and real-time processing, which might be challenging to achieve

with QML without a suitable back-end connection. To address these issues, developers can use a hybrid strategy that combines QML for front-end development with secure, performance-optimized back-end s

1. Quantum Computing:
    - Quantum Bits (Qubits): Unlike classical bits, qubits can exist in multiple states simultaneously, enabling quantum computers to process vast amounts of data much faster.
    - Quantum Algorithms**:** Algorithms like Quantum Support Vector Machines (QSVM), Quantum Neural Networks (QNN), and Quantum K-Means can handle complex computations more efficiently than their classical counterparts.
    - Quantum Annealing**:** Useful for optimization problems, which are common in machine learning, especially in the training of models.
2. Hybrid Machine Learning**:**
    - Classical Machine Learning**:** Techniques like decision trees, logistic regression, and classical neural networks are well understood and can be integrated into the hybrid model.
    - Quantum-classical integration**:** Combining quantum algorithms with classical machine learning methods can enhance the model's performance.
3. Data Integration and Pre-processing**:**
    - Big Data Analytics**:** Handling and processing vast amounts of patient data, including electronic health records (EHRs), medical imaging, and genetic information.
    - IoT Devices**:** Incorporating data from wearable devices and sensors that monitor patients.

ervices. The system utilizes time-series and feature-based classification techniques to ensure a strong early detection mechanism concerning coronary heart disease. Key Components of the Model are discussed below.

Unlike conventional bits, which only represent information as 0 or 1, quantum bits (qubits) can exist in several states simultaneously due to superposition. This enables quantum computers to process massive volumes of data simultaneously, significantly enhancing computational capability for certain challenges. Quantum machine learning (QML) provides considerable benefits in processing multidimensional healthcare data using quantum computing's capacity to conduct complicated computations tenfold quicker than traditional computers. This is especially useful in healthcare, where datasets are generally large and complex, containing patient histories, genetic information, medical imaging, and real-time sensor data. Quantum algorithms, such as quantum-enhanced feature selection or quantum support vector machines, can handle big, complicated information more effectively, revealing patterns and correlations that classical methods may miss. Furthermore, quantum entanglement and superposition enable QML to investigate several data states simultaneously, resulting in more accurate predictions, quicker optimization, and enhanced diagnostic insights, all critical for individualized treatment regimens and early illness diagnosis.

The introduction chapter outlines the background for the research, explaining the problem and the justification for using quantum-assisted machine learning. Chapter 2 provides the foundation for the study, identifies the current research gaps, and positions quantum-enhanced machine learning as a new direction in heart disease prediction. The methodology chapter 3 details the data management specifics, the integration of quantum computing, and the system implementation for real-time predictive analytics. In the results and discussion of Chapter 4, model performances were compared with those obtained. It will also bring forth challenges and indicate the potential directions for improvement. Chapter 5 concludes by summarizing the work done, its importance, and the directions for further research.

2 Literature review

Most of the recent research uses a classic machine learning-based approach toward the early detection of coronary heart disease. This paper provides an overview and analysis of the machine learning classifiers used to diagnose chronic diseases, followed by their implementation and assessment. It illustrates the value of machine learning classifiers in the healthcare industry and how they can produce more accurate predictions. Decision trees, SVMs, random forests, and artificial neural networks are widely used and are trained on data that includes patients'

demographics, past medical history, and test results [12–14]. Olawade et al. [15] and Burton et al. [16] illustrated the efficacy of deep learning algorithms for predicting coronary artery disease using echocardiography data while managing high accuracy related to non-invasive diagnostics. Researchers also use random forests and logistic regression models to estimate CHD based on risk factors such as blood pressure, cholesterol level, diabetes, and other lifestyle parameters [17–19]. Most of them target feature selection methods indirectly to enhance the quality of a prediction with the most minor computational complexity. Since medical datasets are ever-increasing in their dimensionality and complexity, inefficiency regarding computation and interpretation by classical algorithms encourages exploration into quantum enhancement methodologies.

Quantum computing has superior computational abilities and is fast gaining centre stage in machine learning; thus, it has a potential for using machine learning for intricate, high-dimensional data analyses. The quantum computer uses qubits, similar to the bits in classical computers, in that they can solve complicated problems and exchange information [10, 20]. A few research works have started looking at applications of QML in health diagnostics, early cancer detection, and heart-related ailments. According to Sengupta et al. [21], quantum algorithms may significantly speed up the classification tasks related to medical diagnostics. Theoretically, QML outperforms classical algorithms in handling complex and multi-variable interactions from a large dataset, improving accuracy and model speed in disease prediction [22, 23]. However, despite such potential, practical applications of pure QML in the healthcare domain remain in their infancy due to the present-day hardware limitations of quantum computers. Recent work has shown that such hybrid methods can be even more computationally efficient and accurate than both classical models purely and their pure quantum counterparts [24, 25]. Suzuki et al. [26] examine QML models based on quantum support vector classification (QSVC) and regression (QSVR) using a quantum circuit simulator. The performance of QSVC models using 4 qubits of the trapped-ion quantum computer and QSVR models on the near-term quantum device. Rui Zhang et al. [27] demonstrate a QSVM based on the regularized Newton method (RN-QSVM), which achieves an exponential speed-up over the classical algorithm. Havlíček et al. [28] proposed a quantum-enhanced feature space for SVM: quantum kernel estimation. It realized a computational speedup in classification tasks compared to classical SVM. Decoodt P et al. [29] have explored hybrid quantum–classical algorithms for applications in cardiology. Their work demonstrates how it would be possible to embed quantum algorithms in DL models for medical image analysis in an auspicious manner to detect heart diseases. A recent study by Abdulsalam et al. [30] suggests that QML classifiers outperform classical ML classifiers in heart disease prediction. Furthermore, the investigation shows that the bagging ensemble learning method raises the quantum classifiers' prediction accuracy.

The proposed hybrid QML models leveraging both quantum algorithms and the techniques of well-established classical ML can, therefore, offer more accurate and faster predictions, even in complicated and noisy medical datasets. With ongoing improvements in quantum hardware, hybrid models are expected to increase medical diagnostics further. While most of the classical machine learning approaches already demonstrated significant advances in CHD detection, their integration with quantum computing showed a path toward further enhancement. The practical application remains in its infancy, with the development of quantum hardware and algorithms needed to develop hybrid methods in healthcare. Table 1 summarizes a comparative analysis of different machine learning algorithms investigated in this work. This table compares the critical metrics, such as accuracy, precision, recall, F1-score, and computational cost. This comparative framework would highlight the strengths and weaknesses inherent to each algorithm and make it easier to choose the most suitable model for the problem under study.

Classical ML models, particularly deep learning models, demand massive computing resources as data size and model complexity grow. Classical computing fails to scale effectively as data dimensions become huge or when issue complexity exceeds specific thresholds. Quantum algorithms, such as quantum annealing or quantum Fourier transforms, offer exponential speedups for specialized applications like optimization, massive dataset search, and complicated distribution sampling. While quantum algorithms are promising, quantum hardware is still in its early stages. Current quantum computers have limited qubits, error rates, and coherence periods, rendering them unsuitable for large-scale real-world machine learning applications. Classical machine learning approaches frequently fail to address specific quantum challenges, such as modelling quantum systems, optimising across huge quantum states, or dealing with entanglement and superposition principles. So, designing hybrid quantum–classical algorithms that use the capabilities of both quantum and classical systems is not an easy challenge. Classical models are frequently required to handle pre- and post-processing phases.

Table 1 Comparative analysis of different machine learning algorithms

Author/ref. no	Techniques used	RF	NB	GB	SVM	LR	Dataset used	Major findings	Limitations
		Accuracy %							
Kalaparveen Bagadi et al. [31] (2023)	Cardiovascular Disease Prediction Using SVM, gradient boosting, KNN, naive Bayes Classifier, and logistic regression	0.8952	0.9094	0.9096	0.8825	0.926	Long Beach, Va And Stalog (Heart)	Ensemble methods enhanced model robustness and reduced overfitting compared to individual classifiers	Require substantial computational resources for training and tuning, posing challenges for real-time or resource-constrained applications
Hossain et al., [32] (2024)	Machine Learning Approach For Predicting Cardiovascular Disease using Logistic Regression, Naïve Bayes, Decision Tree, AdaBoost, RF, and Bagging Tree	96.15	94.87	94.94		93.67	Random Sample of CVD Patients In Bangladesh	Bagging Tree (BT) improved model stability and reduced variance	Limited model performance for minority classes, requiring advanced resampling techniques. Ensemble models like RF and AdaBoost require significant resources for training and tuning
Khan et al., [33] (2023)	Machine Learning Algorithm-Based CVD prediction using decision tree (DT), RF logistic regression (LR), Naïve Bayes (NB), and SVM	0.8501	0.7474	0.8372	0.8308	0.8303	UCI dataset	Decision Tree (DT) provided good interpretability but was prone to overfitting compared to ensemble methods like RF	Limited dataset size and diversity restricted the generalizability of the results to broader populations. The interpretability of complex models like RF and SVM remains a concern for clinical deployment
Singh et al., [34] (2024)	Heart Disease Detection Using DT, RF, SVM and PCA	0.97			0.6832		Framingham Dataset	Random Forest and PCA proved to be the most effective approach for accurate and efficient heart disease detection	The computational overhead of SVM and the dependency on optimal hyperparameter tuning posed challenges. PCA improved efficiency and caused the loss of some critical features, impacting model interpretability



Fig. 2 The workflow of the proposed QML architecture for CHD detection

3 Methodology

The development and implementation of the quantum-based hybrid ML model for early prediction of cardiac diseases amidst Healthcare 4.0 are indeed complex and multivariate, considering the involvement of sophisticated technologies and methodologies. Advanced data analytics, real-time monitoring, and personalized medicine feature prominently in this era. Indeed, the hybrid quantum–classical ML model is promising in predicting early cardiac disease by exploring both quantum computing and classical algorithms. In machine learning, the QSVM and QNN would see that they can substantially speed up over classical alternatives, particularly when the dimension becomes exceptionally high. Logistic regression, decision trees, and neural networks are a few of the traditional ML models utilized in health care to predict various diseases, including cardiac conditions. Several of these models suffer from high-dimensional data and require increasing computational resources. Some critical limitations of these models, such as feature selection, dimensionality reduction, and optimization, can be overcome by combining quantum computing with the existing models. The flow diagram of the proposed predictive analysis of heart disease using a quantum-assisted machine learning framework is shown in Fig. 2.

The proposed hybrid QML technique, which blends quantum computing's parallelism and exponential speedup capabilities with classical machine learning models, has the potential to improve computational efficiency significantly. Quantum algorithms, such as those for data encoding and optimisation, may process vast volumes of data more quickly than classical algorithms because they use quantum superposition and entanglement, which can handle multidimensional feature spaces more effectively. This allows for quicker training durations, potentially improved accuracy, and better generalisation for diagnosing CHD, particularly in situations involving varied and noisy data. Furthermore, hybrid approaches enable the seamless integration of conventional techniques with quantum algorithms, maximising both classical and quantum resources to overcome traditional ML's inherent scalability and efficiency limitations. QML's low computing cost, particularly in complicated predicting tasks, makes it an attractive option for real-time, large-scale medical applications like CHD diagnosis, where speed, precision, and handling massive datasets are crucial.

3.1 Data collection and pre-processing

The first step involves gathering data from diverse sources. Electronic health records (EHRs) provide comprehensive patient histories, demographic information, lab results, and medical imaging. Data from IoT devices, such as wearable technology, offers real-time monitoring of vital signs like heart rate, blood pressure, and ECG readings. Additionally, genomic data sheds light on genetic predispositions to cardiac diseases, while lifestyle data, including diet, exercise, and smoking habits, further enrich the dataset. Once collected, this data undergoes rigorous pre-processing. Initial steps include data cleaning to remove inconsistencies, handle missing values, and standardize formats. Missing values can be addressed using mean/median/mode imputation, interpolation, or K-Nearest Neighbors (KNN) imputation. Outlier detection is performed using Z-Score calculations or the Interquartile Range (IQR) method to identify and manage anomalies. The data is then normalized or scaled using techniques like Min–Max Scaling, Z-Score Standardization, Robust Scaler, Logarithmic Transformation, or Box-Cox Transformation. Finally, relevant features are extracted to enhance the prediction of cardiac disease.

$$z = \frac{X - \mu}{\sigma} \tag{1}$$

In Eq. 1, μ is the mean, and σ is the standard deviation. The method identifies outliers by measuring the spread of the middle 50% of the data. Data points outside 1.5 × IQR (Interquartile Range) from the first or third quartile are flagged as outliers. An IQR is defined in Eq. 2.

$$IQR = Q3 - Q1 \tag{2}$$

Logarithmic Transformation $X' = \log(X + 1)$, often used to handle skewed data, and X′ is defined in Eq. 3.

$$X' = \frac{X - \text{median}}{IQR} \tag{3}$$

3.2 Quantum-enhanced machine learning

The model's core leverages quantum computing to enhance traditional machine learning algorithms. Quantum Support Vector Machines (QSVM) utilize quantum states to compute kernel functions that measure the similarity between data points, offering a more refined classification process. The QSVM optimization problem involves minimizing a function subject to constraints involving multipliers and data labels. Quantum neural networks (QNN) further advance this by applying unitary transformations to quantum states and optimizing parameters through quantum gradient descent or other quantum optimization techniques. The hybrid approach uses classical methods for initial data pre-processing and feature extraction while employing quantum computing for model optimization and hyperparameter tuning. The iterative training loop integrates both processors, with classical computing handling straightforward tasks and quantum computing addressing more complex ones. Model performance is evaluated using accuracy, precision, recall, and F1-score metrics.

Quantum states compute a kernel function K(xi,xj) that measures the similarity between data points. This can be represented as the inner product of quantum states in Eq. 4.

$$K(xi, xj) = |\langle \phi(xi) | \phi(xj) \rangle| \tag{4}$$

The QSVM optimization problem is typically formulated as:

$$\min_{\alpha} \sum \alpha i \alpha j y i y j K(xi, xj) - \sum_{i} \alpha i \tag{5}$$

In Eq. 9, y_i are the labels, and α_i are the multipliers. Each layer in a QNN applies a unitary transformation U(θ)U(θ) to a quantum state $|\psi\rangle|\psi\rangle$, parameterized by θ as shown in Eq. 6.

$$\theta : \left|\psi'\right\rangle = U(\theta)|\psi\rangle|\psi'\rangle = U(\theta)|\psi\rangle \tag{6}$$

The loss function is typically the expected value of an observable at Eq. 7.

$$L(\theta) = \left\langle \psi'(\theta)|O|\psi'(\theta)\right\rangle L(\theta) = \left\langle \psi'(\theta)|O|\psi'(\theta)\right\rangle \tag{7}$$

3.3 Real-time prediction and monitoring

The model must seamlessly integrate with healthcare systems for practical application. Real-time data ingestion ensures continuous monitoring and updating of patient records from IoT devices. A predictive analytics dashboard is developed to provide healthcare providers with user-friendly visualisations of predictions and risk assessments, facilitating timely interventions and decision-making. Creating a statistical analysis dashboard for real-time prediction and monitoring requires multiple processes. First, relevant data sources are identified and integrated to ensure the information is correct and current. Data is then pre-processed, cleansed, and converted to guarantee consistency and dependability. Advanced predictive models, which frequently include ML or statistical approaches, are used to estimate previous data to estimate future patterns or events. These models are fine-tuned and evaluated to assure accuracy in real-world scenarios.

3.4 Implementation and deployment

The infrastructure setup involves utilizing a hybrid cloud environment, combining classical cloud computing resources with quantum cloud services to support the computational demands of the model. Ensuring data security is paramount, with robust measures such as encryption and access controls implemented to protect patient information. This comprehensive

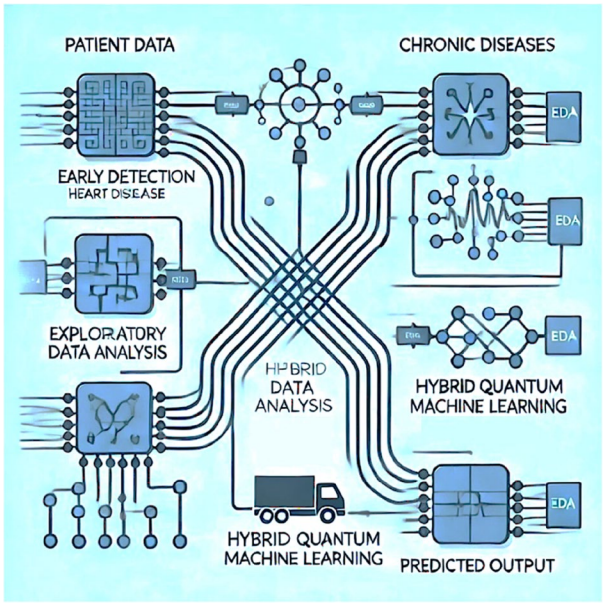


Fig. 3 Block diagram of the proposed hybrid machine learning model using QML architecture

approach ensures the effective deployment of a quantum-based hybrid machine learning model in predicting early cardiac disease and enhancing diagnostic accuracy and patient outcomes. This article analyses the machine learning classifiers used to diagnose chronic diseases, followed by their implementation and assessment. It then illustrates the value of machine learning classifiers in the healthcare industry and how they can produce more accurate predictions. The block diagram for the prediction model of chronic diseases is shown in Fig. 3.

The multi-step inferential framework for integrating quantum and classical machine learning algorithms is a promising approach. Still, its robustness depends on how well the quantum and classical components complement each other within the problem space. Quantum algorithms, particularly those leveraging quantum entanglement and superposition, can offer exponential speedups in optimization and feature selection tasks. Classical algorithms, on the other hand, provide stability, scalability, and a rich array of tools for data pre-processing, model interpretation, and post-processing. Clinical and imaging data utilized in testing often combines patient demographics, such as age, gender, and medical history; clinical measures, such as vital signs, lab findings, and symptoms; and high-resolution imaging data. These datasets attempt to provide a comprehensive picture of patient health across various illnesses. Clinical trials and imaging datasets are frequently designed to adjust for confounding variables and include varied populations, although they may not accurately reflect real-world settings. Factors such as socioeconomic status, comorbidities, and variances in healthcare access can all impact patient outcomes. They may be underrepresented in controlled settings, making these datasets less representative of real-world clinical circumstances.

4 Results and discussion

While choosing the algorithm in the domain of early prediction of cardiac disease using ML, the models' performances remain significant. Each machine learning pipeline includes performance metrics. They provide a progress indicator along with a numerical value. Metrics don't have to be differentiable to track and evaluate a model's performance during testing and training. Metrics like accuracy, precision, F1 score, and recall were used in this study to validate the proposal. The metrics utilised depend on the issues attempted to solve, and each type of indicator gives unique insights into model performance. We have used four evaluation metrics to assess the performance of the quantum-assisted machine learning model in the predictive analysis of heart disease. Precision measures the proportion of correctly predicted positive cases among all positive predictions made by the model. It is defined as:

$$\text{Precision} = \text{True Positives(TP)} / \text{Total Predictions} \tag{8}$$

Recall, also known as sensitivity or true positive rate, evaluates the model's ability to correctly identify all actual positive cases. It is calculated in Eq. 9.

Table 2 Performance analysis of the different classification models

Sl. no	ML model	Accuracy (%)	Precision(%)	Recall(%)	F1 Score(%)
1	KNN	99.99	99.99	100	99.96
2	Random Forest	99.57	99.41	99.72	99.56
3	Linear Regression	99.99	99.99	100	99.99
4	Decision Tree	99.97	99.96	99.97	99.91

Table 3 Performance comparison of the hybrid ensemble model for CHD detection

Sl. no	Classification model	Accuracy(%)	Precision(%)	Recall(%)	F1 (%)score	AUC	Cohens kappa
1	CNN	99.88	99.87	99.89	99.88	0.985	0.9423
2	Random Forest	98.60	98.61	98.60	98.61	0.998	0.9758
3	Ensemble 1(CNN + RF)	99.88	99.87	99.90	99.88	0.988	0.9758
4	KNN-RF Ensemble	99.96	99.95	99.97	99.96	0.99	0.99
5	DT-RF Ensemble	99.93	99.93	99.92	99.92	0.99	0.978
6	LR-RF Ensemble	98.79	98.81	98.73	98.77	0.988	0.975
7	AdaBoost-RF Ensemble	99.93	99.94	99.92	99.93	0.999	0.998
8	Gradient Boost-RF Ensemble	98.79	98.81	98.73	98.77	0.974	0.860

$$\text{Recall} = \text{True Positives(TP)} / (\text{True Positives(TP)} + \text{False Negatives(FN)}) \tag{9}$$

The F1 Score is the harmonic mean of precision and recall, providing a balanced measure of the model's performance, especially in scenarios involving imbalanced datasets. It is expressed in Eq. 10.

$$\text{F1 Score} = 2 * (\text{Precision} * \text{Recall}) / (\text{Precision} + \text{Recall}) \tag{10}$$

Accuracy assesses the model's performance by accounting for correct positive and negative predictions. It is represented in Eq. 11 below.

$$\text{Accuracy} = [\text{TP} + \text{TN}] / [\text{TP} + \text{TN} + \text{FP} + \text{FN}] \tag{11}$$

Equations 8–11 evaluated the quantum-assisted machine learning model's effectiveness in predicting heart disease, ensuring a thorough analysis of its precision, recall, and overall accuracy.

Table 2 summarises important metrics of four various ML models: KNN, Random Forest, Linear Regression, and Decision Tree. Each has been evaluated for accuracy, precision, recall, F1 score, Cohen's Kappa score, and AUC. In this hybrid approach, we take advantage of the stacking notion; a meta-classifier was paired with several classification models as an ensemble approach known as stacking. Multiple layers were arranged one after the other, with the models fulfilling their predictions in each layer. The top layer model is the base model, which informs choices by combining many models. The original data provides properties for the models in the lower layer. The model's top layer receives input from the lower levels and uses it to give a final prediction. To process the original data, the stacking approach uses numerous independent machine-learning models as input. Subsequently, the weights of each machine learning model are computed, and the input and output are predicted using the meta classifier. The algorithms with the best performance are chosen, while those with poor performance are eliminated. This method uses a meta-classifier to train many machine learning algorithms on the same dataset after combining numerous classifiers as the basis model. Table 3 shows the hybrid algorithms for cardiovascular data. Each has been evaluated for accuracy, precision, recall, F1 score, Cohen's Kappa score, and AUC.

In Table 3, the hybrid QML models outperform traditional ML models in terms of accuracy, sensitivity, F1 score, and specificity for coronary heart disease. QML models use quantum computing principles, such as superposition and entanglement, to rapidly process and analyze complicated data, resulting in greater generalization and classification accuracy. They can outperform traditional models such as Support Vector Machines (SVM), Decision Trees, and Random Forests, particularly on high-dimensional datasets. While classical models may excel in some areas, QML

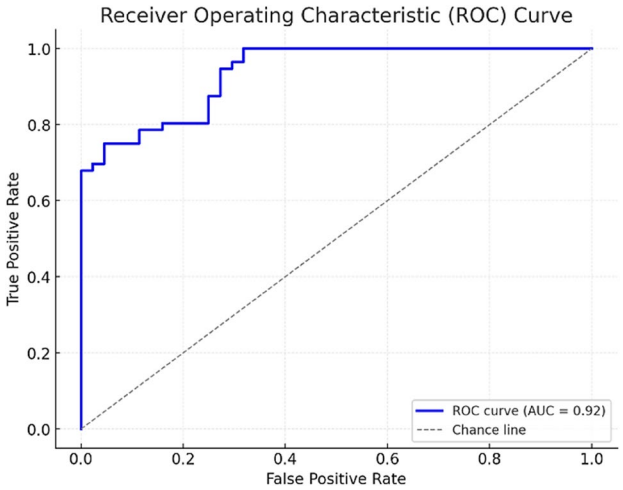


Fig. 4 AUC Graph of hybrid ensemble model for Coronary heart disease detection

models frequently achieve a superior balance across all metrics, providing high accuracy, greater sensitivity, better specificity, and a higher F1 score.

CNN shows quite good results in all metrics, especially with a high recall of 99.89, which means that it captures nearly all positive cases. That makes this model reliable for early detection but slightly outperforms some ensemble models. Random Forest performs solidly, though less accurate than CNN and ensemble methods. While it has consistent performance in overall accuracy and precision, it lags. This model matches CNN's accuracy but has a higher recall at 99.90, making it marginally better at detecting true positive cases. The hybrid K-Nearest Neighbors and Random Forest) The ensemble model is the strongest, with the highest accuracy of 99.96%, precision of 99.95%, recall of 99.97%, and F1 score of 99.96%. AdaBoost-RF Ensemble performs very well, with accuracy and precision close to the top. It is an outstanding choice, with overall performance just slightly worse than KNN-RF. Similar to the LR-RF ensemble performance, the Gradient Boost-RF Ensemble model has good performance but not as high as other ensembles. This makes it less suitable for choice. The other ensembles, LR-RF and Gradient Boost-RF, turn out only averagely and are less suitable, given that they yield lower accuracy and F1 scores. Finally, hybrid ensemble models, notably the KNN-RF Ensemble, stand out as very effective solutions for early detection of coronary heart diseases compared to solo models such as CNN and traditional Random Forest.

Figure 4 shows the AUC Graph of the hybrid ensemble model for Coronary heart disease detection. A High AUC (0.92) demonstrates strong discriminative ability, effectively differentiating between the two classes. The curve stays well above the diagonal, confirming that the model performs significantly better than random guessing. AUC values above 0.9 generally indicate that the model is highly reliable and could be used confidently in decision-making processes.

In Fig. 5, the confusion matrices presented for the six ensemble models demonstrate their exceptional performance in binary classification tasks. All models exhibit high precision and recall with minimal misclassifications. KNN-RF, AdaBoost-RF, and Gradient Boost-RF Ensemble models stand out for their near-perfect classification, characterized by extremely low false positive and false negative rates, indicating high sensitivity and specificity. While all models can correctly identify positive and negative instances, KNN-RF achieved the highest performance, with only 1 false negative and 2 false positives. These results suggest that the ensemble approaches effectively combine the strengths of individual classifiers, delivering reliable and accurate predictions across various classification challenges.

This research develops a hybrid machine-learning prediction model for chronic diseases in such a manner that it ensures the optimum level of accuracy. It is observed that the same dataset with different ML methods has differences in its accuracy scores to make a better model. While using cross-validation to solve the overfitting problem and reach the highest accuracy by combining decision trees, gradient boosting, Gaussian naive Bayes, and gradient boosting, our primary focus will remain the selection of the best hybrid model. The future of quantum computing in Healthcare 4.0, driven by machine learning algorithms, will massively influence medical science and patient care because it reveals unprecedented data analysis and processing capability. Quantum algorithms will allow data from different sources in real time, including wearable devices, to support proactive health management and dynamic decision-making. Moreover, integrating quantum computing with machine learning will drive innovations in areas such as genomic research, whereby capabilities to model and analyze the genetic data at the quantum level will deliver insights more profoundly from mechanisms of diseases to targets for therapies.

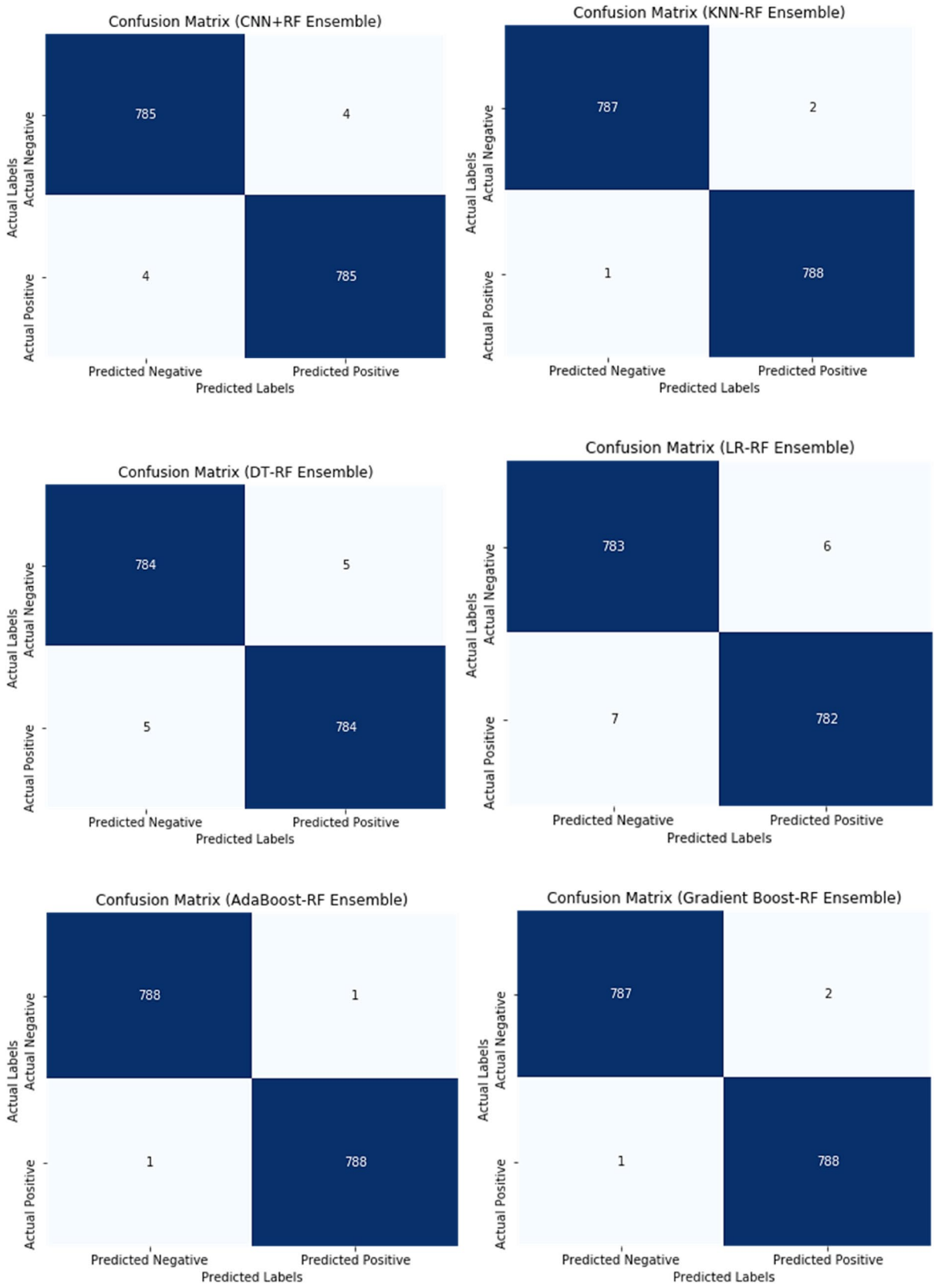


Fig. 5 Confusion matrix of CNN + RF, KNN-RF, DT-RF, LR-RF, AdaBoost-RF and Gradient Boost-RF Ensemble models

Hybrid QML models may successfully handle noise and unpredictability in healthcare information by combining classical and quantum computing approaches. A hybrid model's quantum component may process vast, complicated datasets in ways that classical models alone cannot. This enables the hybrid QML model to better detect underlying patterns, even in noisy or incomplete data. Furthermore, quantum algorithms like quantum classifiers or quantum support vector machines can improve model resilience by recognizing critical characteristics and lowering the impact of irrelevant noise. The proposed strategy increases predicted accuracy and makes it possible to find intricate patterns in patient data that conventional approaches would miss by skillfully fusing the advantages of several algorithms. In managing chronic diseases, where differences in each patient's health profile can significantly impact the effectiveness of treatment, this degree of personalization is essential. Our results also show that this hybrid strategy has the potential to be a dynamic clinical decision-support tool. This model can improve proactive care tactics and patient outcomes by continuously learning from fresh data and adjusting to changing health patterns. Our strategy increases predicted accuracy and makes it possible to find intricate patterns in patient data that conventional approaches would miss by fully fusing the advantages of several algorithms, such as supervised classifiers and unsupervised clustering techniques. The research results also demonstrate how this hybrid technique may be used as a dynamic decision-support tool in clinical situations. This model can improve proactive treatment approaches and patient outcomes by constantly learning from fresh data and adjusting to new health trends. Beyond its immediate clinical uses, this finding has implications that could radically change how chronic diseases are managed. Improved machine learning techniques can help healthcare systems transition to a more personalized and predictive model of treatment that not only addresses current health issues but also looks ahead to future needs.

The Raspberry Pi 5 GPU platform plays a pivotal role in developing and deploying the proposed QAML model for the predictive analysis of heart disease. The Raspberry Pi 5's GPU capabilities facilitate parallel processing, accelerating computationally intensive tasks such as matrix operations and training processes in ML models. The compact size and energy-efficient architecture of the Raspberry Pi 5 make it suitable for deploying the trained predictive model in edge-computing scenarios, such as in medical devices or remote healthcare systems. The platform's affordability and modularity ensure scalability by allowing the deployment of multiple units across healthcare centers or research facilities. It serves as a bridge between resource-constrained environments and quantum computing systems by offloading pre-processing and intermediate computations. Table 4 summarizes the hardware and software characteristics of Raspberry Pi 4B used during this study. The parameters outlined in the table include the important technical parameters in this area, such as the kind of processor, amount of memory, storage options, and implementation environment.

Due to its versatility, computational capabilities, and economical efficiency, the Raspberry Pi 4B is ideal for predicting heart disease through quantum-assisted machine learning. Since the Raspberry Pi cannot execute the native quantum algorithms, it is good for executing classical machine learning. It can connect with cloud-based quantum computing platforms for more complex quantum-assisted functionalities. This setup is ideal for real-time healthcare applications and cost-effective experimentation with advanced machine learning techniques, combining edge processing with quantum cloud computing on the Raspberry Pi.

The suggested QML architecture, while initially created for specific disorders, has great adaptability to a wide variety of chronic and acute diseases, demonstrating its versatility. QML's capacity to handle complicated, high-dimensional data might be used to model illness development, predict therapy responses, and uncover biomarkers for diabetes, cardiovascular disease, neurodegenerative disorders, and even cancer. Its ability to handle a wide range of datasets, from genetic information to clinical records, allows the framework to be adapted to specific illness profiles, giving useful insights for personalized treatment and improving clinical outcomes across several medical domains.

5 Conclusion

Quantum algorithms, such as quantum support vector machines and quantum neural networks, promise improved predictive accuracy by exploring extensive solution spaces and capturing intricate patterns that classical models might miss, facilitating earlier and more precise disease detection. The performance for early detection of coronary heart disease is best for the KNN-RF Ensemble and DT-RF Ensemble; KNN-RF is the best overall model, with the highest accuracy of 99.96% and a recall of 99.97%, making it very good at finding true positives with minimal false negatives, which is very important in medical diagnosis. It is also impressively performed by the AdaBoost-RF Ensemble and DT-RF Ensemble. It is only slightly worse than the KNN-RF ensemble and thus can be regarded as a feasible alternative for early detection

Table 4 Hardware and Software Specifications of Raspberry Pi 4B used in this research

Component	Specifications
Hardware	
Raspberry Pi 4B Model	Raspberry Pi 4 Model B (4 GB or 8 GB RAM variant)
Processor	Quad-core ARM Cortex-A72 CPU, 1.5 GHz
Graphics	Broadcom VideoCore VI GPU, supporting OpenGL ES 3.0
Memory (RAM)	8 GB LPDDR4-3200 SDRAM
Storage	microSD card (32 GB)
Power Supply	5 V 2A, USB-C power supply
Operating System	Linux-based Raspberry Pi OS
Software	
Programming Languages	Python
Machine Learning Libraries	TensorFlow, Scikit-learn, and Keras
Quantum Computing SDK	IBM Qiskit
Data Science Libraries	Pandas, NumPy and SciPy
Machine Learning Models	Random Forest, SVM, Decision Trees, KNN, Neural Networks, Quantum Machine Learning algorithms (QML)
Visualization Tools	Matplotlib and Seaborn

tasks. As quantum technology advances, its integration into healthcare machine learning will drive transformative changes, unlocking new research possibilities, optimizing treatments, and setting new medical innovation and patient care standards. This research developed a hybrid quantum machine-learning prediction model for chronic renal disease that prioritizes accuracy. To create a better model, we have examined the accuracy of the same dataset using several machine learning methods and compared their accuracy scores. This research uses cross-validation to solve the overfitting problem and achieve the highest accuracy when combining popular machine learning classifiers like decision trees, gradient boosting, Gaussian Naïve Bayes, and gradient boosting to create the best hybrid model. This study shows a promising hybrid machine-learning technique for managing and detecting chronic diseases early on. Our model efficiently leverages the intricacy of patient data by combining several supervised and unsupervised machine learning techniques, increasing prediction accuracy and enabling tailored therapies. The results highlight how crucial it is to have a thorough analytical framework that can handle a variety of data sources and capture complex patient characteristics. To improve the model's responsiveness and enrich the dataset, future work will integrate patient-reported outcomes and extend this model to include real-time data from wearable devices.

Acknowledgements Our journal manuscript, "Predictive Analysis of Heart Disease Using Quantum-Assisted Machine Learning", is completed, and we would like to take this opportunity to thank everyone who helped make it possible. First and foremost, we would like to express our gratitude to the esteemed corresponding authors of this research, Dr Sherin Zafar, Prof. Dr Parul Agarwal, Prof. Dr M Afshar Alam, Dr Siddhartha Sankar Biswas, Dr Imran Hussain and Dr Abubeker K M. Their outstanding leadership, unmatched knowledge, and unshakable commitment have been crucial to our study project at every turn. We sincerely appreciate the kind assistance from FIST-DST (Department of Science and Technology—Foundation for Infrastructure and Sustainable Technologies). Their support has been crucial in helping us to complete this project. Their invaluable support made the work reported in this publication feasible by giving us the infrastructure and technical tools needed to conduct the research efficiently and precisely. We especially want to express our sincere gratitude to the anonymous reviewers whose perceptive comments, helpful recommendations, and helpful criticism have greatly improved the calibre and readability of this work. Their knowledge and astute criticism have been extremely helpful in raising the overall calibre of our study. Finally, we would like to express our profound gratitude to all the people and organisations that have helped with this project.

Author contributions Mehroush Banday: Conceptualization and design of the model, development of the theoretical foundation of the model, formulation of the research hypothesis, evaluation of the system's performance, drafting and revising of the manuscript. Sherin Zafar: Led the conceptualization and design of the quantum-assisted machine learning model for heart disease prediction. Developed the theoretical foundation of the model, formulated the research hypothesis, and helped draft the initial manuscript. Parul Agarwal and M Afshar Alam: Evaluated the system's performance, refined the model during the testing phase, and actively contributed to the manuscript development and the critical review process. Siddhartha Sankar Biswas and Imran Hussain contributed to drafting and revising the manuscript. Abubeker K M: Played a central role in implementing and optimizing the quantum-enhanced machine learning algorithms, including integrating quantum components into the classical machine learning models.

Funding No Funding.

Data availability The data supporting this study's findings were collected from the Local Pathology Lab in Muzaffarnagar, Uttar Pradesh, India. It is also provided within the manuscript and will be shared at the request of the Corresponding Author.

Declarations

Ethics approval and consent to participate Not Applicable.

Consent for publication Not Applicable.

Competing Interests The authors declare no competing interests.

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